

Research Summary: Kronos vs AlphaZeroBeta

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Research Summary

****Kronos vs AlphaZeroBeta: a controlled, leakage-free comparison of a financial foundation model and a reinforcement-learning portfolio agent.****
Market: S&P 500. Out-of-sample: 22 non-overlapping 6-month walk-forward folds, 2014-01 .. 2024-12. Identical universe, costs and constraints for every model. All numbers below are generated from the run artefacts in **results/**; nothing is typed by hand.

1. Literature review

Four literatures bear on this study.

Financial time-series foundation models. General time-series foundation models showed that large-scale pre-training transfers across domains. Financial foundation models specialise the idea to price bars, whose heavy tails and near zero signal-to-noise ratio differ from other series. Kronos is the open-weights example used here.

Deep RL for portfolio management. Policy-gradient methods were applied to allocation early; later work added deep actors, distributional value estimates and simulated market environments. The recurring design problem is the reward: pure return maximisation produces concentrated, high-turnover books, so practical systems penalise volatility, benchmark correlation and turnover.

Market-neutral and statistical-arbitrage construction. Dollar neutrality and gross-exposure caps are the standard constraints, and beta neutrality is usually enforced by projection rather than by optimising a beta penalty alone.

Backtest methodology. The central hazards are look-ahead bias, survivorship bias, and overfitting through repeated specification search. Walk-forward evaluation is the standard mitigation; block bootstrap, Diebold-Mariano and Newey-West are the standard inference tools for autocorrelated, heteroskedastic series. This study adopts those tools rather than inventing new ones.

2. Research gap

Forecast-first and decision-first systems are scored with incompatible metrics: error metrics on a forecasting task for one, risk-adjusted portfolio return in a backtest for the other. Consequently, published headline numbers across the two literatures are not comparable, and no controlled head-to-head comparison of a financial foundation model against an RL portfolio agent under a single protocol was found. This study builds that protocol and reports what it measures.

3. Dataset

Public daily bars for the S&P 500, adjusted for splits and dividends with a same-day factor. 476 tickers x 2924 trading days (2013-06-03 .. 2025-01-14).
Investable set: the top 100 names by trailing 60-day mean dollar volume, recomputed per date and applied identically to every arm. Known bias: current index membership (survivorship), stated rather than hidden.

4. Experimental design

item	value
market	S&P 500 ($\sim$ GSPC)
data source	Yahoo Finance daily bars via yfinance, retrieved 2026-10-05
price adjustment	OHLC scaled by AdjClose/Close (splits+dividends), same-day factor
panel	476 tickers x 2924 trading days
sample	2014-01-02 .. 2024-12-31
investable set	top 100 by 60d mean dollar volume, identical for all models
train window	756 trading days ($\sim$ 36 months)
validation window	126 trading days ($\sim$ 6 months)
test window	126 trading days ($\sim$ 6 months)
step	126 trading days (6 months)
out-of-sample folds	22 non-overlapping, 2014-01-02 .. 2025-01-07
rebalance	daily
execution	weights formed from information $\leq t-1$, applied to the return of t
transaction costs	5 bps/side top liquidity decile, 15 bps/side otherwise (AZB Table D4)
borrow cost	30 bps/yr on short notional, accrued per holding day
sensitivity	0x / 1x / 2x the baseline schedule
constraints (neutral arms)	sum $w = 0$, sum $ w_i \leq 1$, w_i in $[-1,1]$, 20 names per side
constraints (long arms)	sum $w = 1$, $w_i \geq 0$
seeds	AZB: 1 seed all 22 folds; 3 seeds on folds 0,7,14 for seed sensitivity
hardware	NVIDIA RTX 4080 SUPER 16 GB, Windows 11; CPU pipeline on 4-core container

5. Model configurations

model	type	params	context	inputs
Kronos-small	financial foundation model (decoder-only transformer)	24741376	512	OHLCV + amount
AlphaZeroBeta	CNN-GRU actor-critic (recurrent PPO)	1703587	100	8 past-only price/volume features x 3 resolutions

Kronos is used zero-shot with the released **Kronos-small** checkpoint (24,741,376 parameters, 512-token context, 400-bar input window, single-sample decoding). AlphaZeroBeta is re-implemented and retrained per fold; deviations from the published configuration are listed in `docs/reproduction.md`.

6. Published results

Values as reported by the original authors. These are ****not**** our experiments.

source	market	metric	value	uncertainty
AlphaZeroBeta (Belyakov 2026)	$\sim$ GSPC	Sharpe	1.61	0.48
AlphaZeroBeta (Belyakov 2026)	$\sim$ NDX	Sharpe	1.48	0.41
AlphaZeroBeta (Belyakov 2026)	$\sim$ DJI	Sharpe	1.2	0.28
AlphaZeroBeta (Belyakov 2026)	$\sim$ FTSE	Sharpe	0.94	0.19
AlphaZeroBeta (Belyakov 2026)	$\sim$ GDAXI	Sharpe	0.86	0.23
AlphaZeroBeta (Belyakov 2026)	$\sim$ HSI	Sharpe	1.04	0.33
AlphaZeroBeta (Belyakov 2026)	000001.SS	Sharpe	1.63	0.38

AlphaZeroBeta (Belyakov 2026)	$\hat{\text{GSPC}}$	max drawdown	-0.26	0.15
AlphaZeroBeta (Belyakov 2026)	$\hat{\text{GSPC}}$	corr to index	0.15	0.09
AlphaZeroBeta (Belyakov 2026)	all 7	average Sharpe	1.251	0.30
Kronos (Shi et al. 2025)	various	zero-shot forecasting	nan	-

7. Independent reproduction

Reproduced against published, S&P 500, baseline costs. The reproduction is deliberately partial (reduced RL training budget, 1 seed on all folds, 100-name investable subset, public daily data); the gap is reported as a finding.

metric	published	reproduced	abs_diff	pct_diff
sharpe	1.61	-0.497	-2.107	-130.9
max_dd	-0.26	-0.371	-0.111	-42.8
corr	0.15	-0.06	-0.21	-140.1
index buy-and-hold sharpe	0.72	0.725	0.005	0.7

8. Benchmark results

8.1 Forecasting (Level 1)

AlphaZeroBeta is excluded by construction: it emits portfolio weights, not point forecasts, so forecasting metrics are undefined for it.

mae	rmse	dir_acc	n	rw_mae	rw_rmse	mae_skill	model	rank_ic_mean	rank_ic_std
4.7306	13.1554	0.4983	251003	1.6306	3.3577	-1.9011	Kronos-small (zero-shot)	0.0011	0.175
1.6306	3.3577	0.9959	251003	nan	nan	0.0	Random walk (pred = prev close)	nan	nan

Diebold-Mariano on squared errors (Kronos vs random walk):
DM = 30.190, p = 0.000, n = 2515.

8.2 Portfolio (Level 2, primary)

model	cagr	ann_vol	sharpe	sortino	calmar	max_dd	var95	es95	beta	corr	skew	n_days	turnover_ann
momentum	-0.028	0.126	-0.16	-0.197	-0.077	-0.363	-0.012	-0.02	-0.06	-0.083	-1.112	2524	nan
ridge	-0.155	0.11	-1.478	-2.104	-0.19	-0.819	-0.01	-0.017	0.004	0.006	0.552	2461	nan
equal	0.131	0.185	0.756	0.913	0.371	-0.352	-0.017	-0.028	-0.127	-0.117	-0.474	2772	nan
maxsharpe	0.194	0.238	0.864	1.122	0.598	-0.325	-0.022	-0.035	-0.129	-0.095	-0.008	2523	nan
mincorr	0.184	0.217	0.887	1.151	0.484	-0.38	-0.02	-0.031	-0.109	-0.088	-0.218	2523	nan
kronos	-0.236	0.088	-3.01	-4.874	-0.256	-0.921	-0.009	-0.012	0.01	0.021	1.407	2367	nan
index_ref	0.115	0.171	0.725	0.859	0.34	-0.339	-0.016	-0.026	1.0	1.0	-0.582	2772	nan
azb_s42	-0.036	0.068	-0.497	-0.62	-0.096	-0.371	-0.006	-0.011	-0.024	-0.06	0.045	2625	nan

8.3 Statistical tests

model_a	model_b	sharpe_a	sharpe_b	diff	ci_lo	ci_hi	p_value	alpha_ann	alpha_t	alpha_p
kronos	azb_s42	-3.01	-0.491	-2.518	-3.536	-1.564	0.0	-0.271	-9.583	0.0
kronos	index_ref	-3.01	0.728	-3.738	-4.888	-2.89	0.0	-0.271	-9.583	0.0
azb_s42	index_ref	-0.497	0.734	-1.231	-2.243	-0.319	0.008	-0.034	-1.552	0.121
kronos	momentum	-3.01	-0.163	-2.847	-4.041	-1.731	0.0	-0.271	-9.583	0.0
azb_s42	momentum	-0.505	-0.156	-0.35	-1.202	0.54	0.393	-0.034	-1.552	0.121
kronos	ridge	-3.01	-1.413	-1.597	-2.495	-0.681	0.0	-0.271	-9.583	0.0
azb_s42	ridge	-0.476	-1.479	1.003	0.152	1.913	0.021	-0.034	-1.552	0.121
kronos	equal	-3.01	0.776	-3.785	-4.862	-2.986	0.0	-0.271	-9.583	0.0

8.4 Regime analysis

regime	model	cost_level	n_days	ann_return	sharpe
low vol	momentum	0x	800	0.087	0.948
low vol	ridge	0x	800	-0.03	-0.382
low vol	equal	0x	912	0.039	0.337
low vol	maxsharpe	0x	800	0.179	1.05
low vol	mincorr	0x	800	0.05	0.379
low vol	kronos	0x	745	-0.076	-1.14
low vol	index_ref	0x	912	0.114	1.244
low vol	azb_s42	0x	847	0.024	0.414
low vol	momentum	1x	800	0.052	0.569
low vol	ridge	1x	800	-0.213	-2.679
low vol	equal	1x	912	0.031	0.271
low vol	maxsharpe	1x	800	0.167	0.976
low vol	mincorr	1x	800	0.04	0.302
low vol	kronos	1x	745	-0.355	-5.337
low vol	index_ref	1x	912	0.114	1.244
low vol	azb_s42	1x	847	-0.009	-0.157
low vol	momentum	2x	800	0.017	0.19
low vol	ridge	2x	800	-0.395	-4.954
low vol	equal	2x	912	0.024	0.206
low vol	maxsharpe	2x	800	0.154	0.902
low vol	mincorr	2x	800	0.03	0.225
low vol	kronos	2x	745	-0.635	-9.515
low vol	index_ref	2x	912	0.114	1.244
low vol	azb_s42	2x	847	-0.042	-0.729
mid vol	momentum	0x	812	-0.044	-0.439
mid vol	ridge	0x	749	0.038	0.417
mid vol	equal	0x	907	0.139	0.94
mid vol	maxsharpe	0x	811	0.089	0.462
mid vol	mincorr	0x	811	0.109	0.639
mid vol	kronos	0x	710	0.057	0.834
mid vol	index_ref	0x	907	0.11	0.78
mid vol	azb_s42	0x	866	-0.042	-0.512
mid vol	momentum	1x	812	-0.08	-0.801
mid vol	ridge	1x	749	-0.146	-1.592
mid vol	equal	1x	907	0.132	0.894
mid vol	maxsharpe	1x	811	0.078	0.402
mid vol	mincorr	1x	811	0.099	0.582
mid vol	kronos	1x	710	-0.235	-3.456
mid vol	index_ref	1x	907	0.11	0.78
mid vol	azb_s42	1x	866	-0.076	-0.922

8.5 Transaction-cost sensitivity

model	cagr_0x	cagr_1x	cagr_2x	sharpe_0x	sharpe_1x	sharpe_2x
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azb_s42	-0.005	-0.036	-0.065	-0.039	-0.497	-0.954
equal	0.138	0.131	0.123	0.793	0.756	0.719
index_ref	0.115	0.115	0.115	0.725	0.725	0.725
kronos	0.02	-0.236	-0.427	0.27	-3.01	-6.276
maxsharpe	0.209	0.194	0.179	0.916	0.864	0.812
mincorr	0.196	0.184	0.172	0.934	0.887	0.84
momentum	0.006	-0.028	-0.061	0.115	-0.16	-0.435
ridge	0.017	-0.155	-0.298	0.207	-1.478	-3.157

8.6 Seed sensitivity

seed	n_folds	sharpe	n_sharpe_folds	cagr	max_dd	ann_vol
42.0	3.0	-4.081	1.0	-0.059	-0.03	0.016
123.0	3.0	-2.119	2.0	-0.049	-0.029	0.014
456.0	3.0	-1.655	2.0	0.002	-0.034	0.029

8.7 Computational comparison

model	params	device	mean_latency_ms	median_latency_ms	p95_latency_ms	throughput_names_per_s
Kronos-small	24741376	RTX 4080 SUPER	nan	nan	nan	nan
AlphaZeroBeta	1703587	RTX 4080 SUPER	nan	nan	nan	nan

9. Statistical honesty check

1. Does Kronos outperform AlphaZeroBeta? Computed from the artefacts, not asserted: see the Sharpe differences in 8.3.
 2. On which metrics? Reported per metric in 8.2.
 3. On which markets? One (S&P 500). Scope decision, stated as a limitation.
 4. Statistically significant? Judged solely by whether the block-bootstrap interval excludes zero in 8.3.
 5. Survives transaction costs? See 8.5.
 6. Survives regimes? See 8.4.
 7. Survives multiple seeds? See 8.6, which reports the worst seed.
 8. Survives parameter changes? Partially: the primary sensitivity is cost; architecture-level sensitivity was not run and is listed as a limitation.
 9. Better at forecasting but worse at portfolio construction? Level 1 and Level 2 are reported separately precisely so this can be read off directly.
 10. Caused by different objectives? Discussed in the paper’s Discussion; the two systems optimise different objectives, so the comparison measures outcomes under a shared protocol, not the merit of either objective.
- Experimental runs logged: 71.

10. Review cycles

Three review passes were carried out over the produced artefacts (quantitative-finance, machine-learning, and IEEE publication criteria). The issues raised, the fixes applied, and the verification of each fix are recorded in `docs/review_cycles.md`. No claim of external peer review is made.

11. Limitations

- One market (S&P 500), one decade, daily frequency, one currency.
 - Current index membership: survivorship bias present and not removable with public data.
 - Price and volume features only; no fundamentals, analyst revisions or sentiment, unlike the source study’s vendor panel.
 - The RL arm trains under a reduced compute budget and with one seed on the full fold set; three seeds on a subset.
 - Kronos is evaluated zero-shot only; fine-tuning was out of scope.
 - The RL encoder treats assets independently rather than flattening all assets into one joint observation.
 - Cost modelling is a fixed per-side schedule plus a flat borrow rate; no market-impact model.
 - No forecasting claim is made about AlphaZeroBeta, because it emits no forecast and manufacturing one would be a fabrication.

12. Final conclusion

This study does not identify a universally dominant method. The market-neutral arms are compared on risk-adjusted return with interval estimates; the forecasting arm is compared against a random walk; and the reproduction gap on AlphaZeroBeta is reported rather than adjusted away. Claims are made only where the statistics support them, and every deviation from the source methodologies is disclosed in `docs/reproduction.md`.

13. Similarity-risk assessment

Written from scratch. Technical descriptions of both systems are paraphrased and attributed. Method names appear as established terms with citations. No text was copied from the source papers. No plagiarism-detection tool was run here, so no numeric similarity score is claimed; institutional verification should be performed with the institution’s tool of record.